Project-Execution of a Basic Quantum Circuit on the IBM Quantum Platform

Problem Statement

To set up and execute a basic quantum computing program using the IBM Quantum platform. The task involves accessing the IBM Cloud environment, creating the required API credentials, running a quantum circuit through a Jupyter Notebook in Google Colab, and observing its execution on real quantum hardware.

Objectives-

1. To understand how to access and use the IBM Quantum platform through IBM Cloud.
2. To generate and configure an API key and instance credentials required for quantum computing experiments.
3. To run a sample quantum circuit using a Jupyter Notebook in Google Colab.
4. To execute the quantum circuit on real quantum hardware and analyze the job details and execution status on the IBM Quantum platform

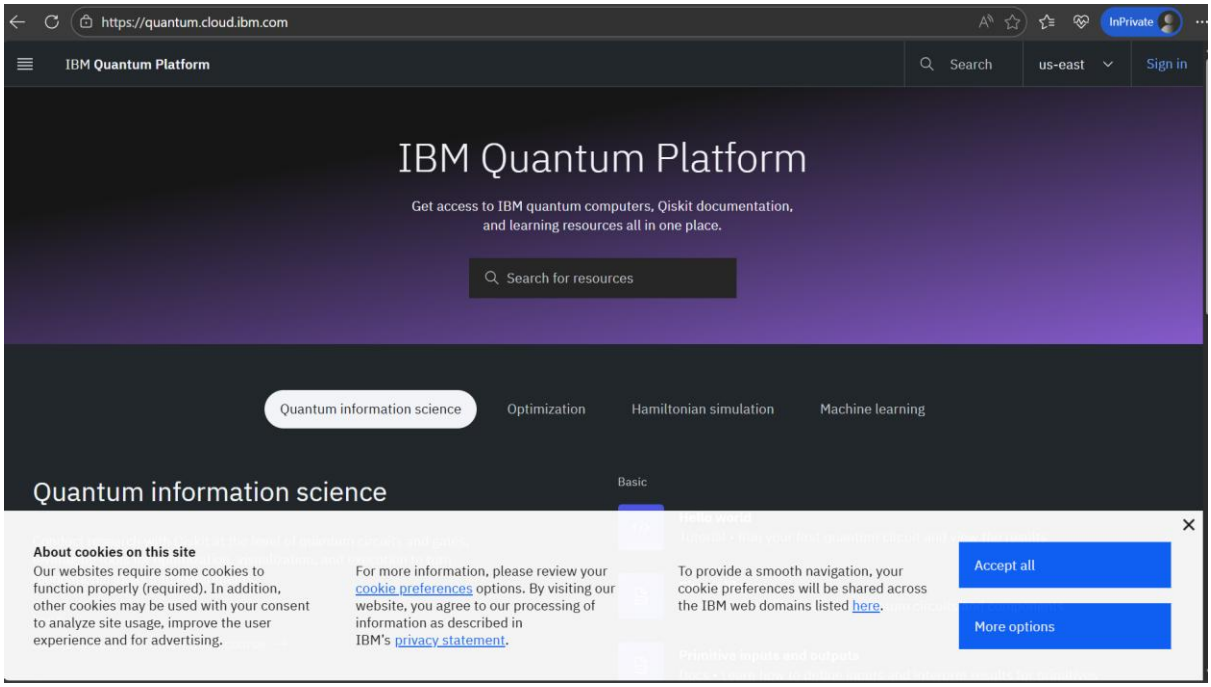
Instructions - Steps by Step Guide-

Pre-requisites

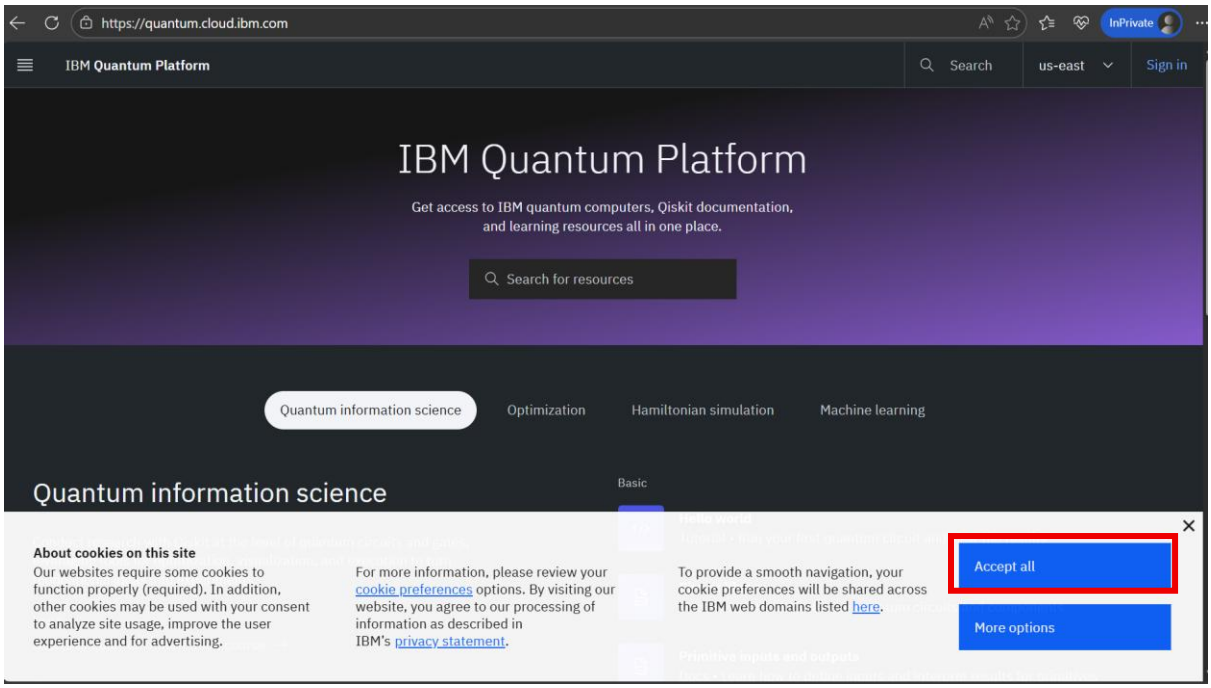
- IBM Cloud Credentials

Procedure

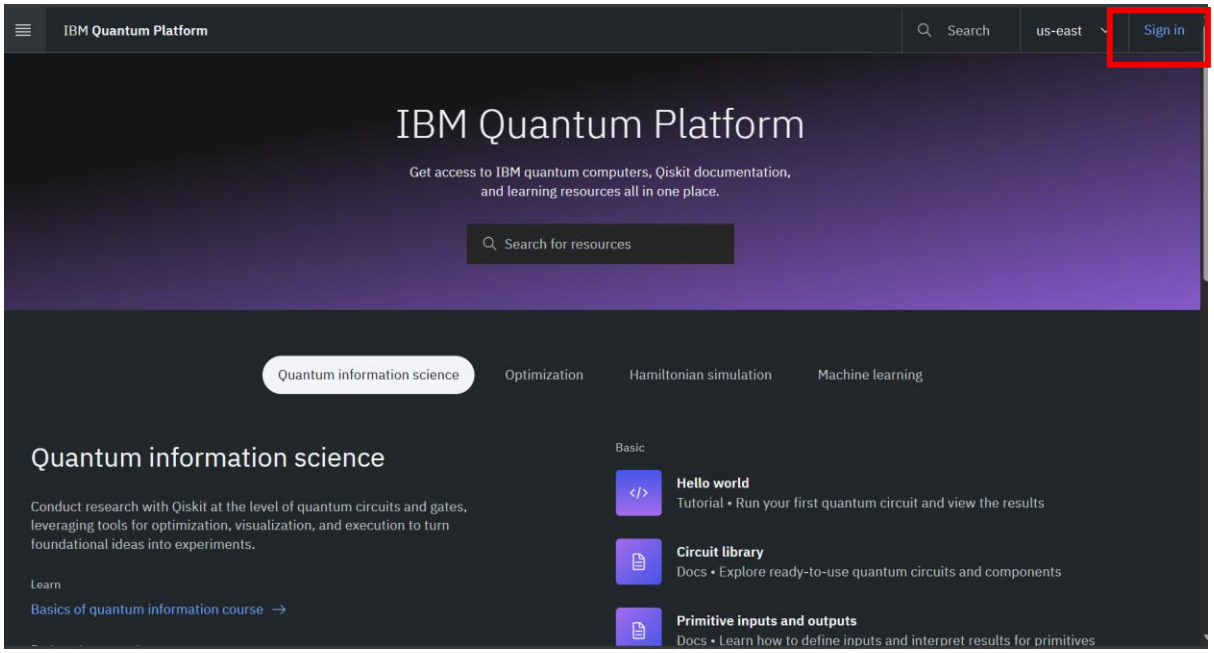
1. Enter the URL - https://quantum.cloud.ibm.com/



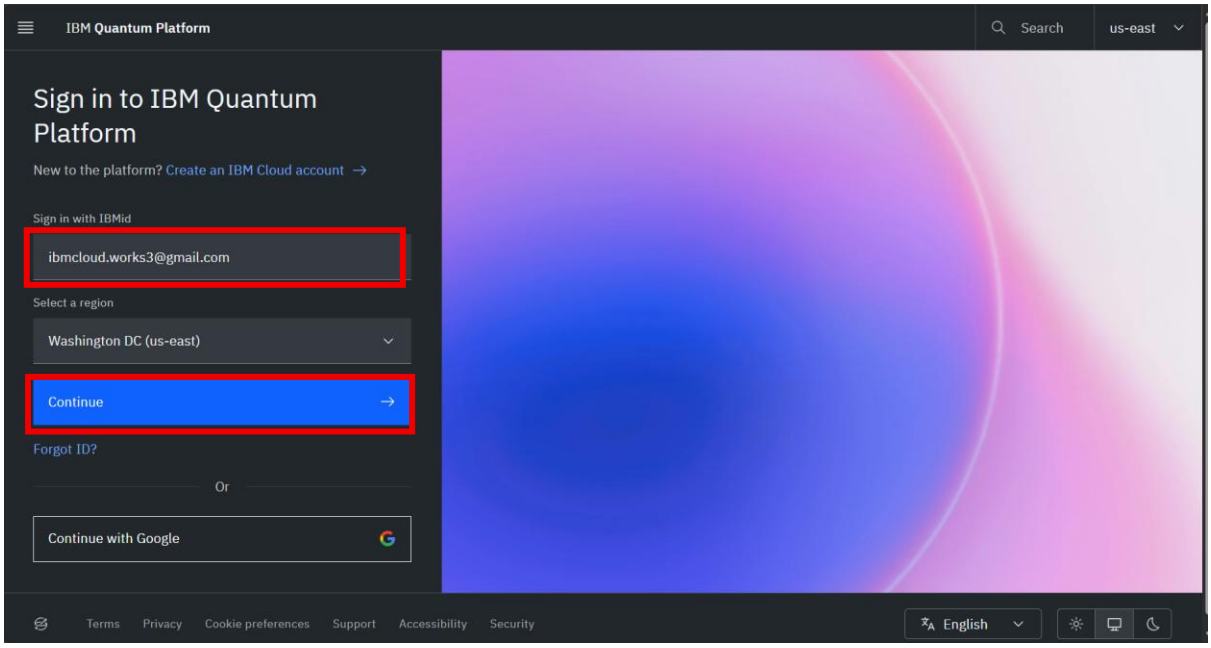
2. Click on Accept All button



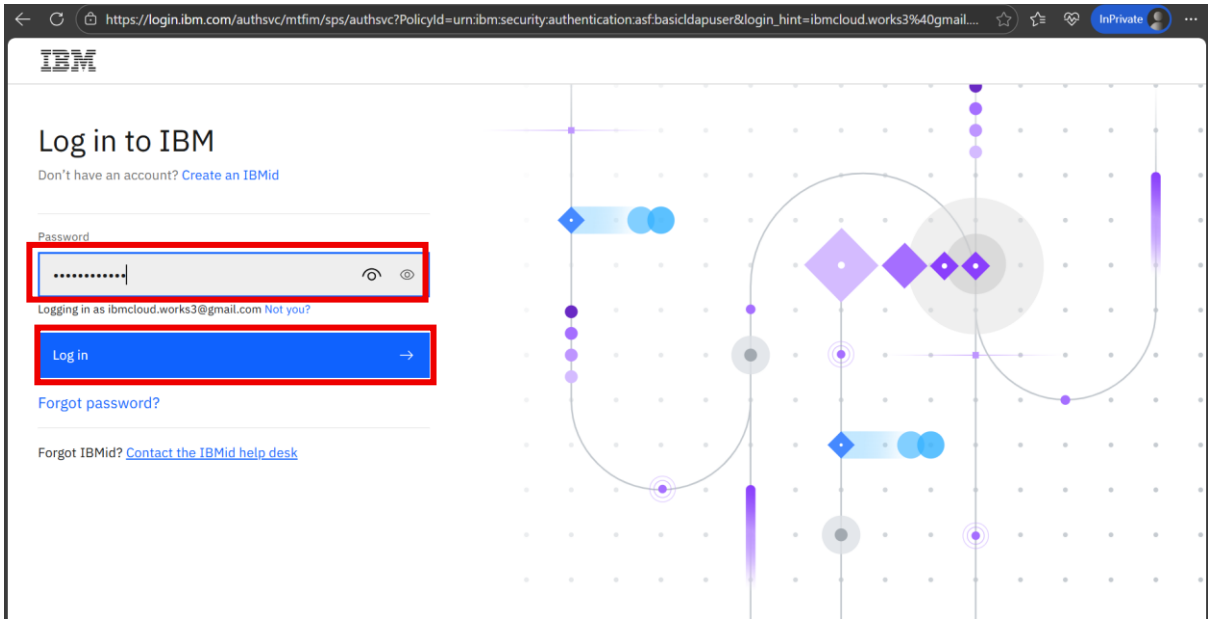
3. Click on Sign In option on the top right corner



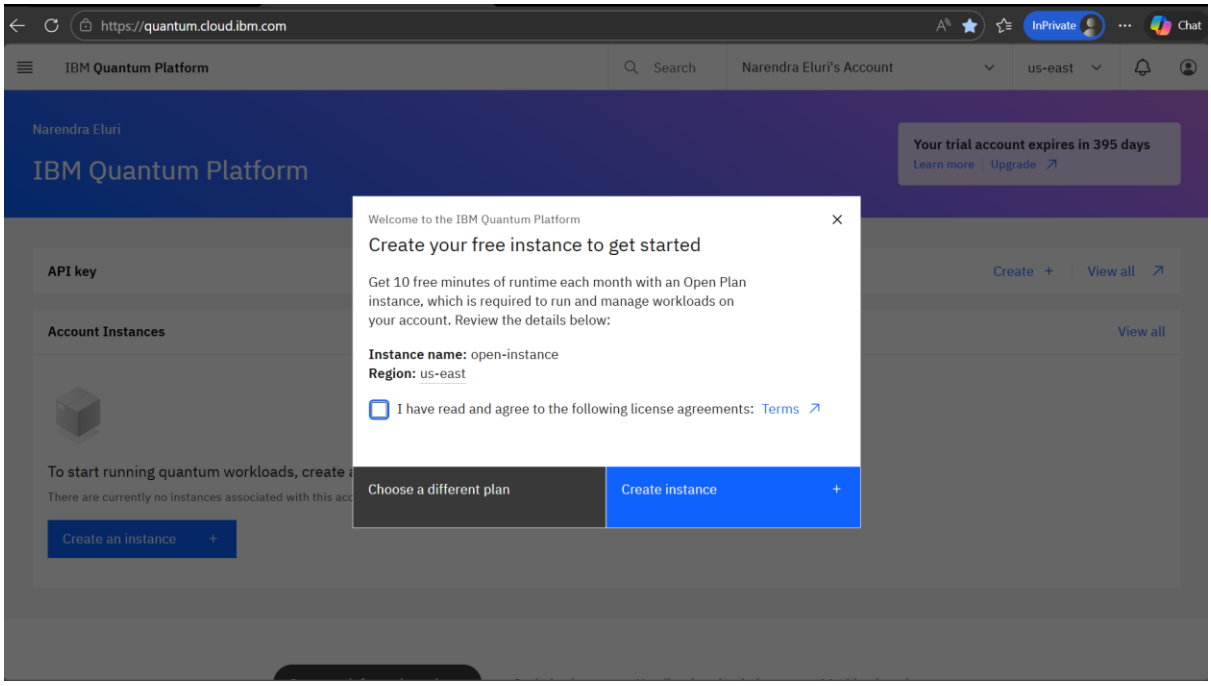
4. In the following page, enter your IBMID, **do not change the region**, click on Continue button



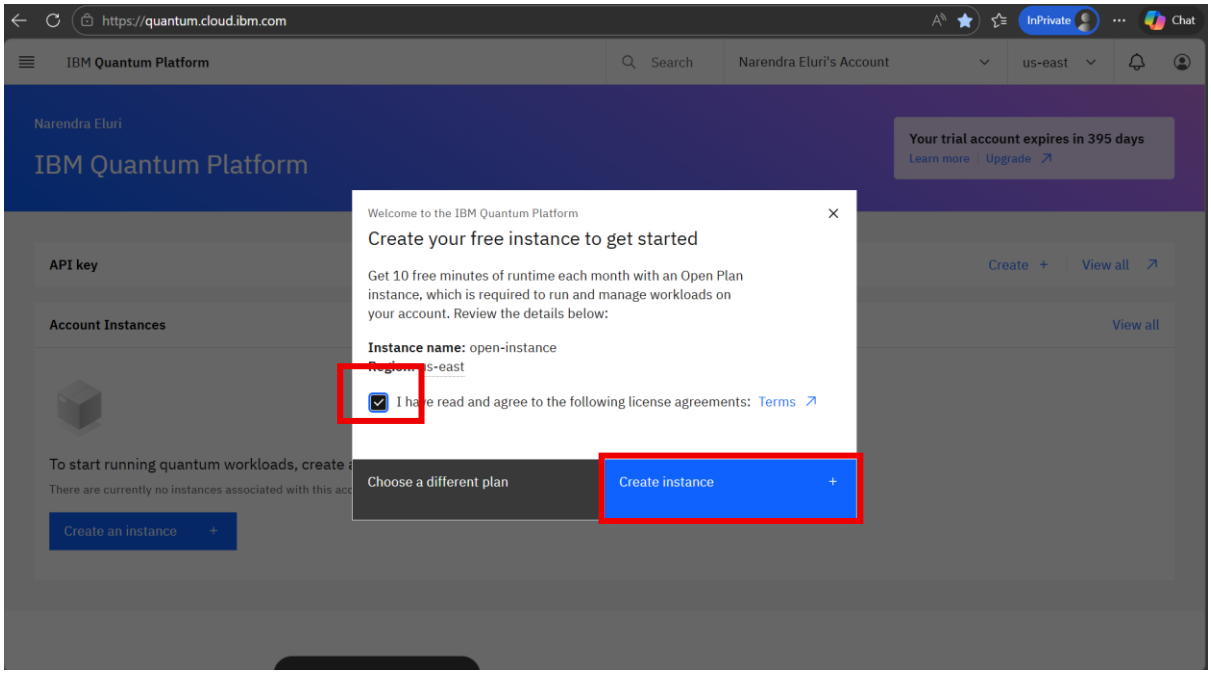
5. Enter the password of your IBM Cloud and click on Log in button



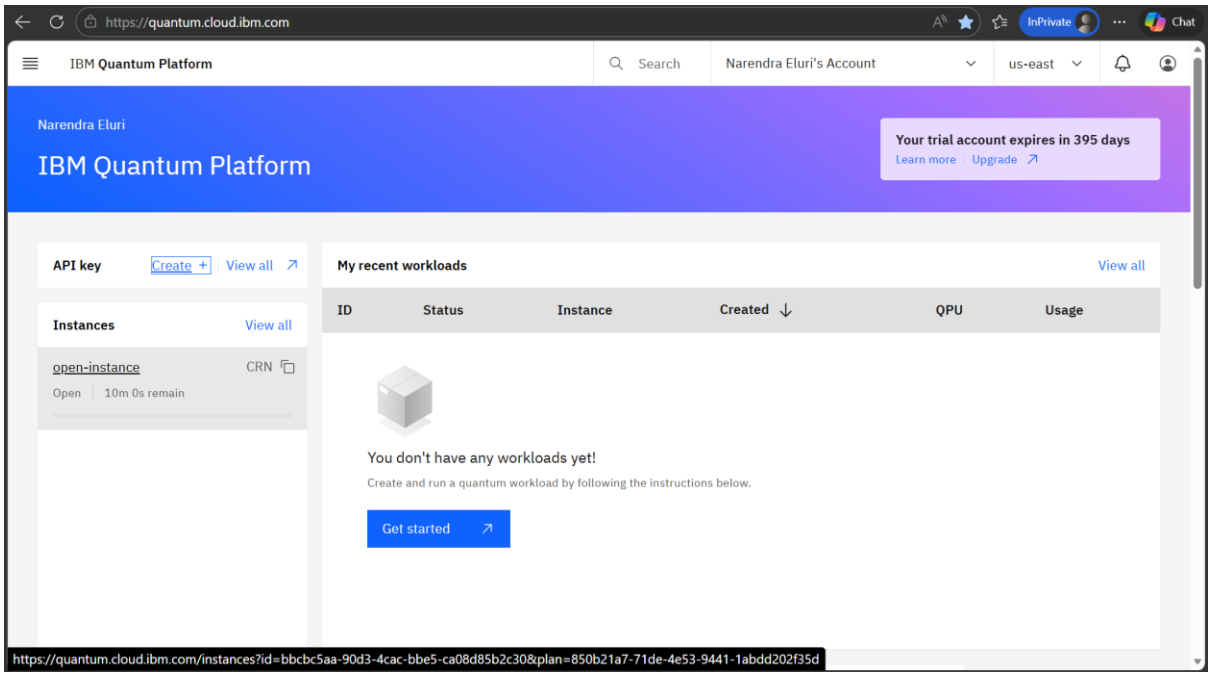
6. The interface will look as below



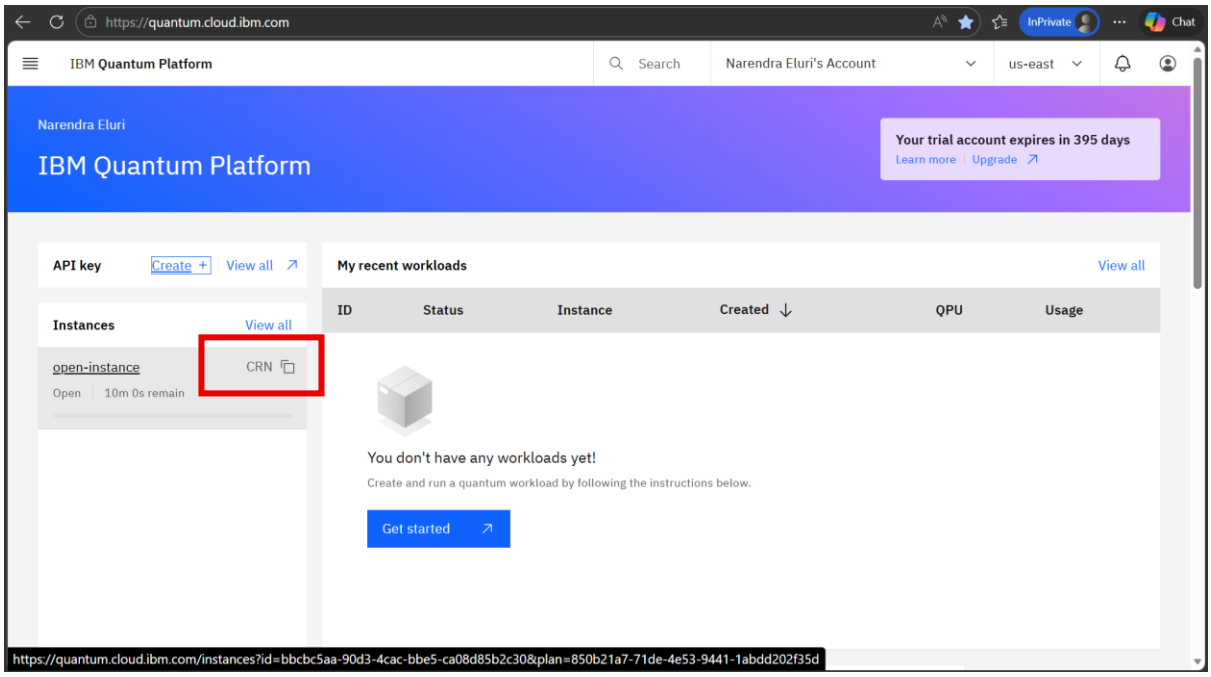
7. Tick the checkbox and click on create instance button



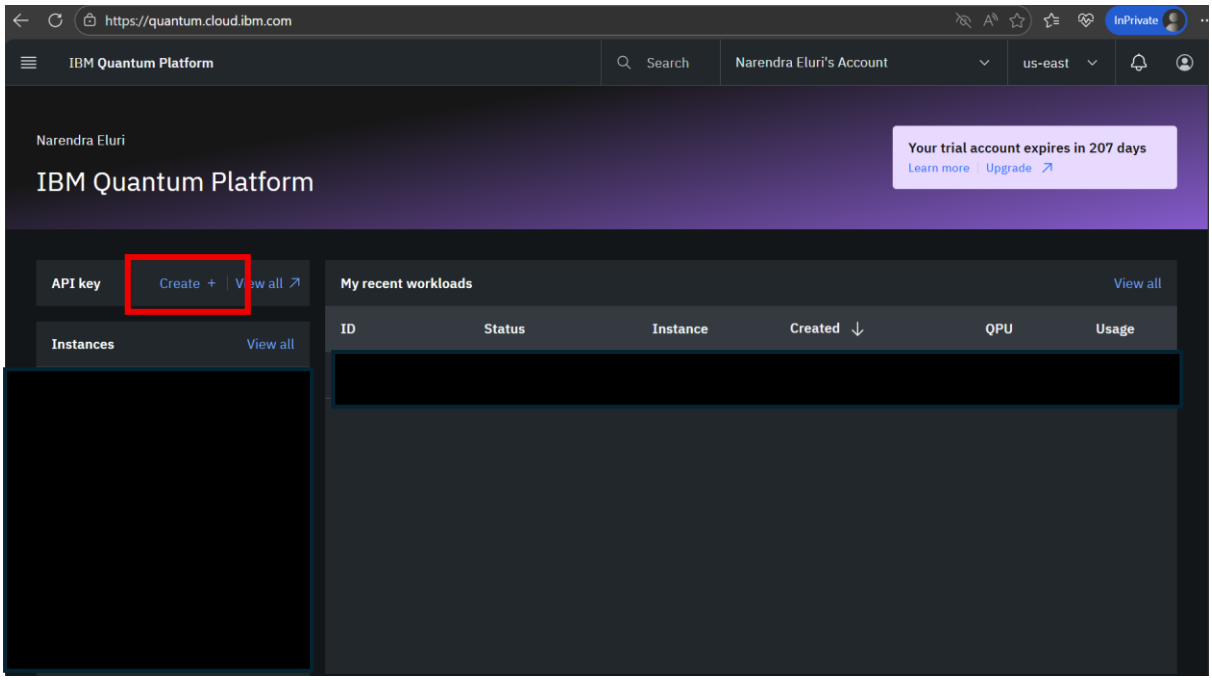
8. Now, the interface is looks as below



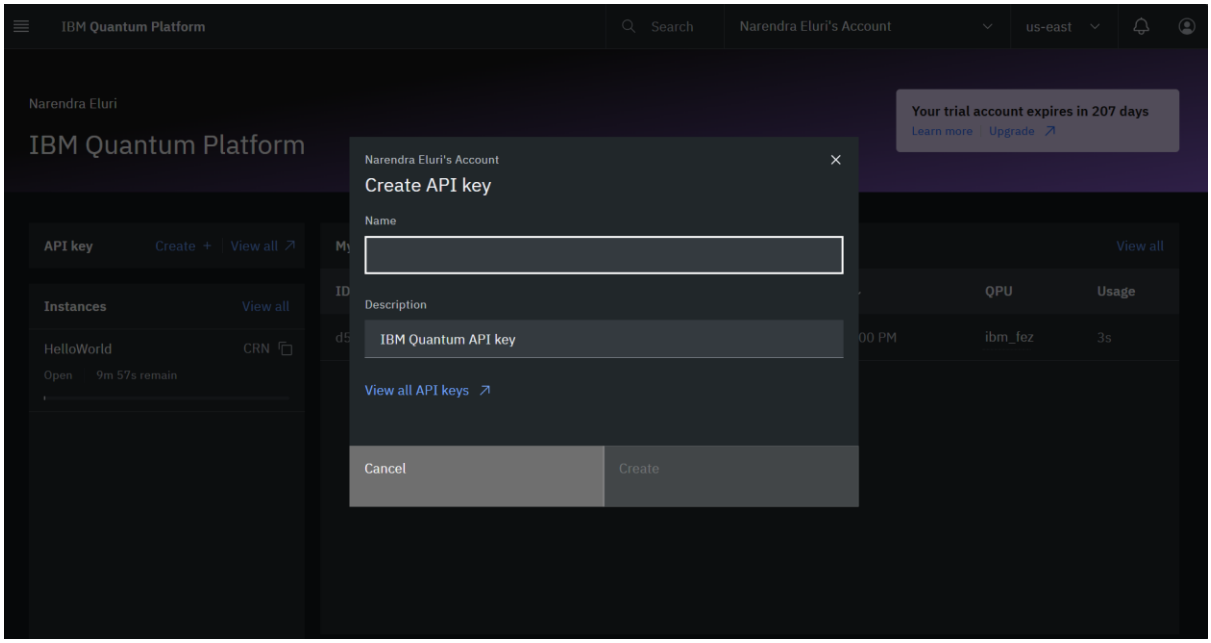
9. Copy the CRN and save the CRN on your notepad



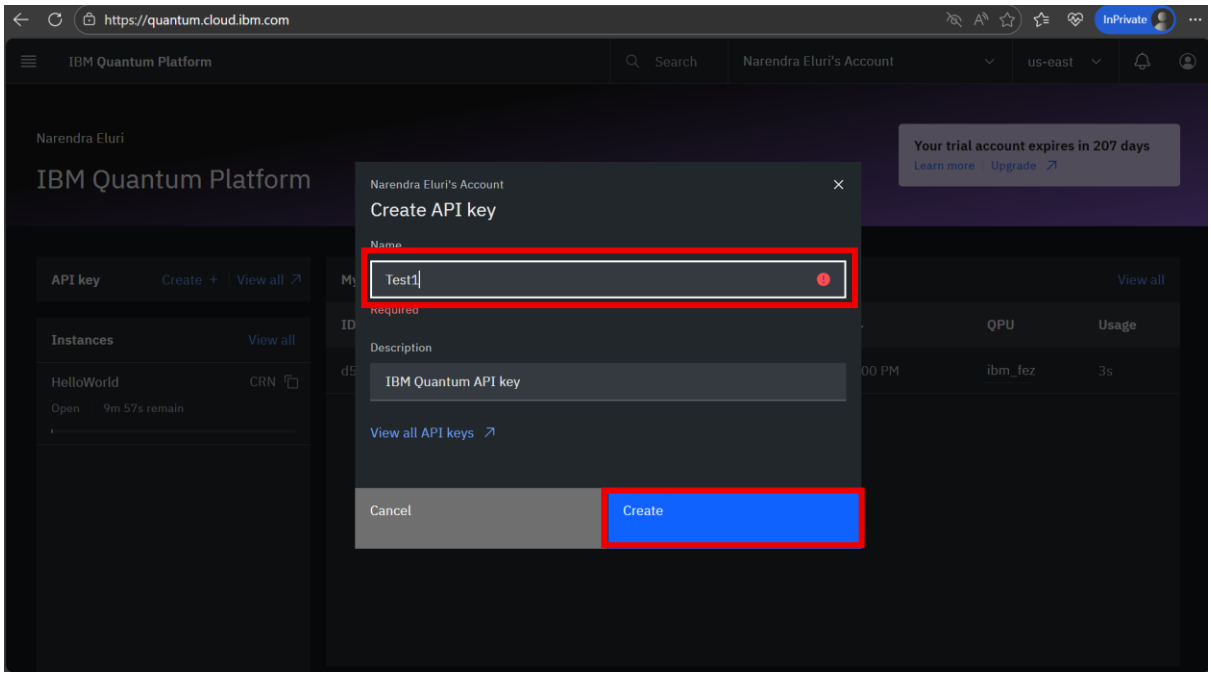
10. Now, Click on **create** to create a new API Key



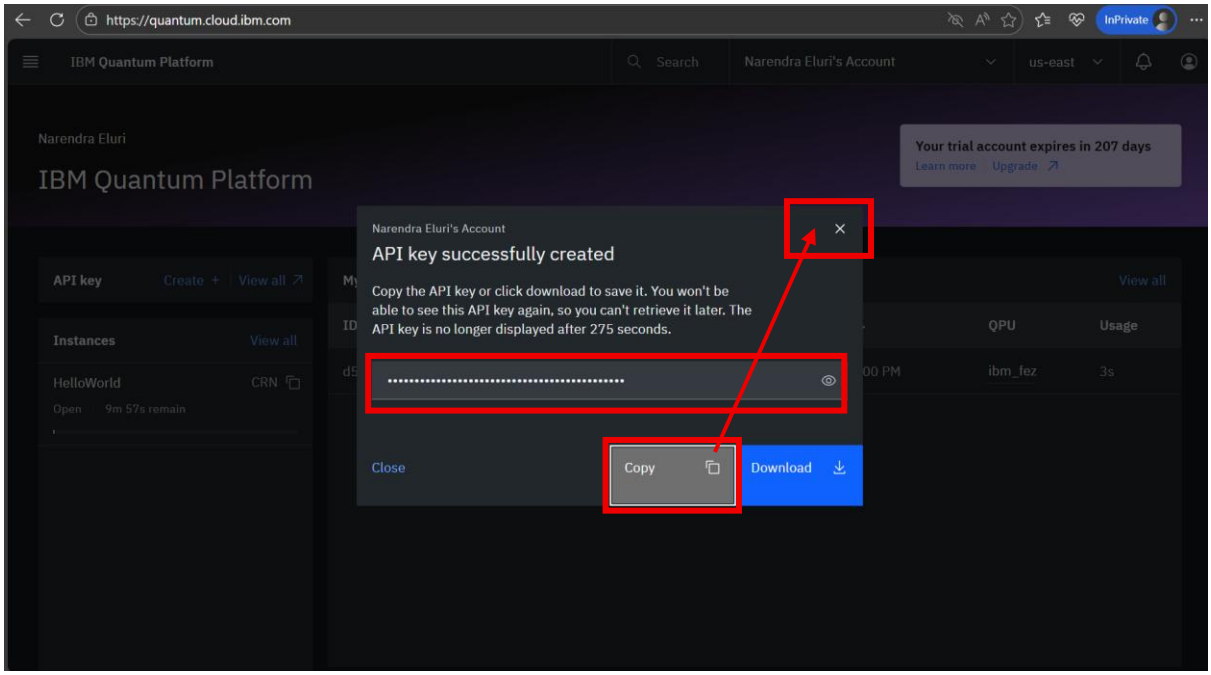
11. Create API Key window popup



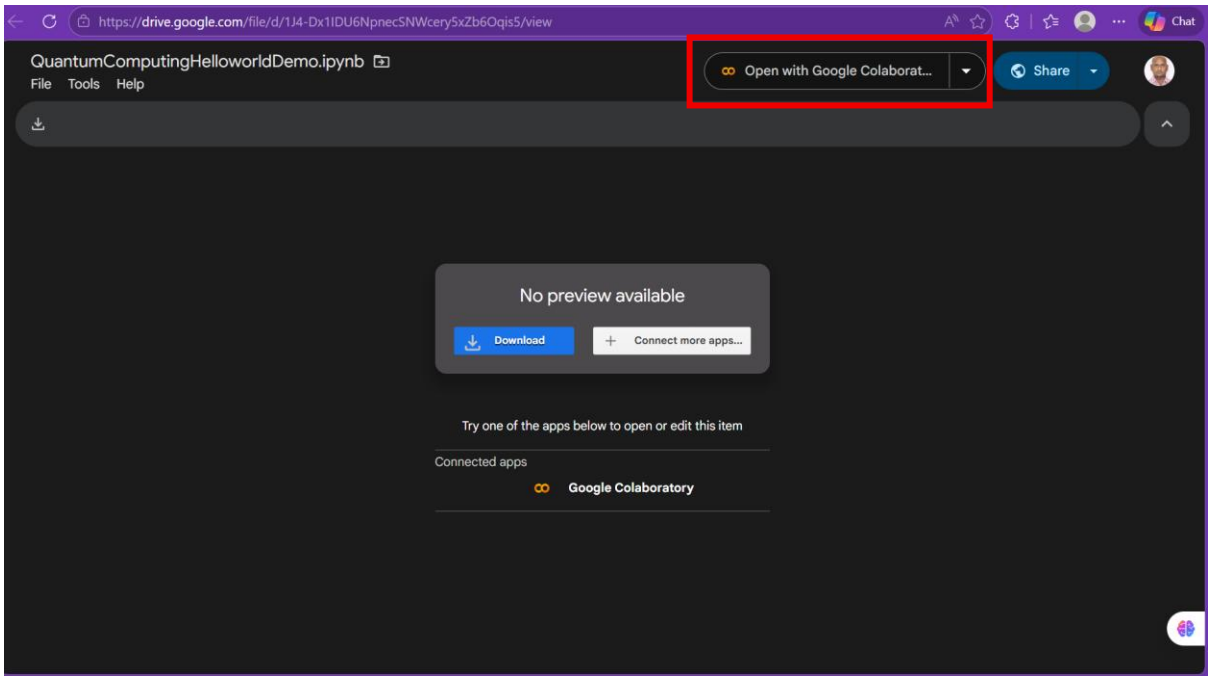
12. Enter Name of the API Key and Click on create button



13. New API Key is created, copy it and save it in notepad and close the window



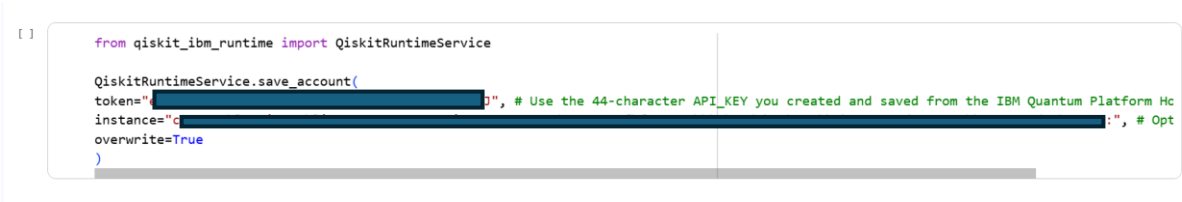
14. Now, click on the link provided below and click on Open with Google Colaboratory option



Link:

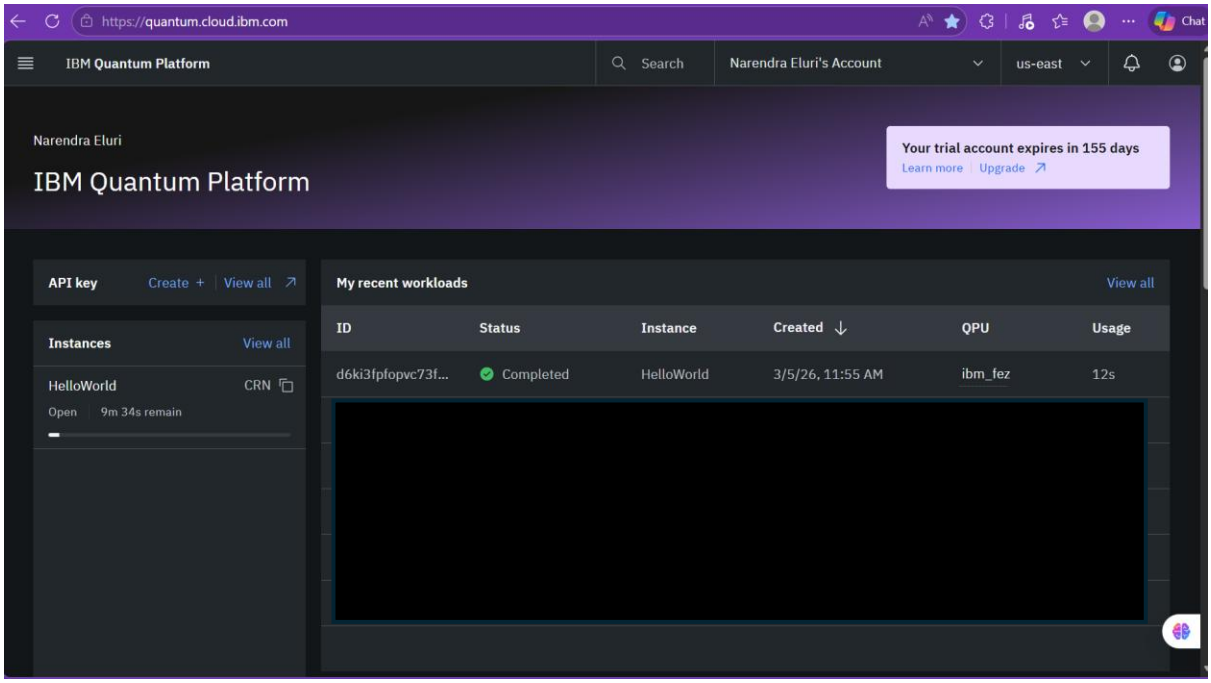
https://drive.google.com/file/d/1vH3ZC_f-tIKRge-XvxQYDE6vHGrcMs5V/view?usp=sharing

15. Go to 3rd cell and replace the token with Your API Key, instance with your CRN and save

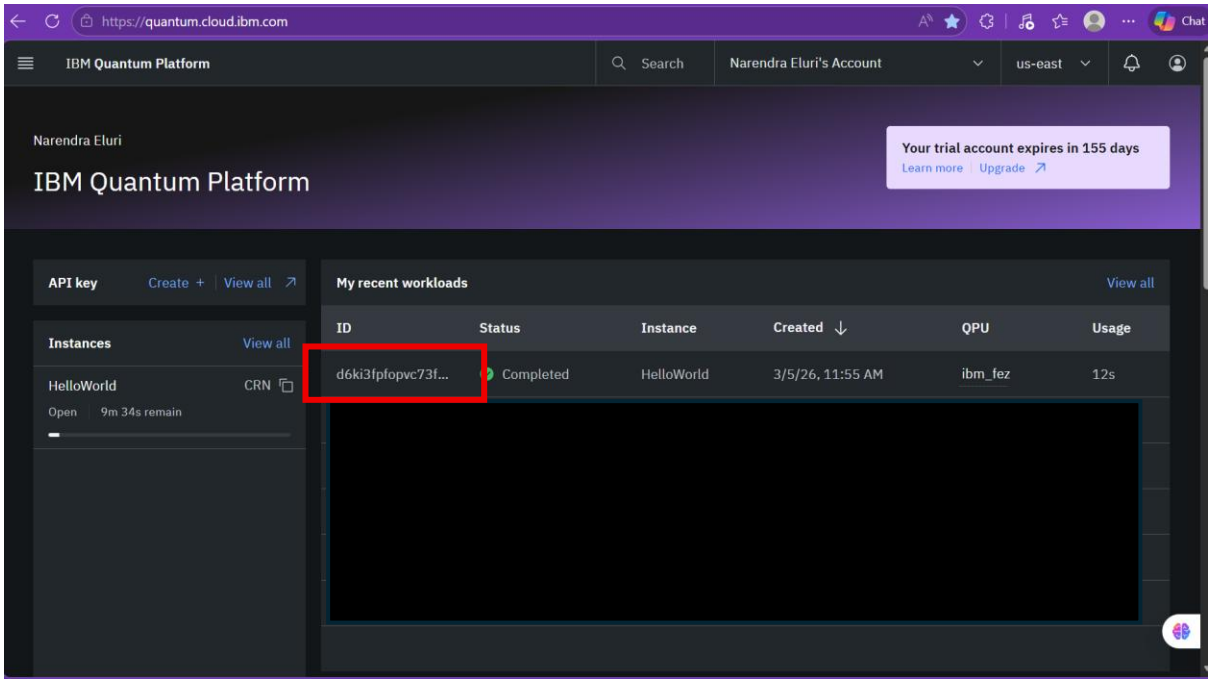


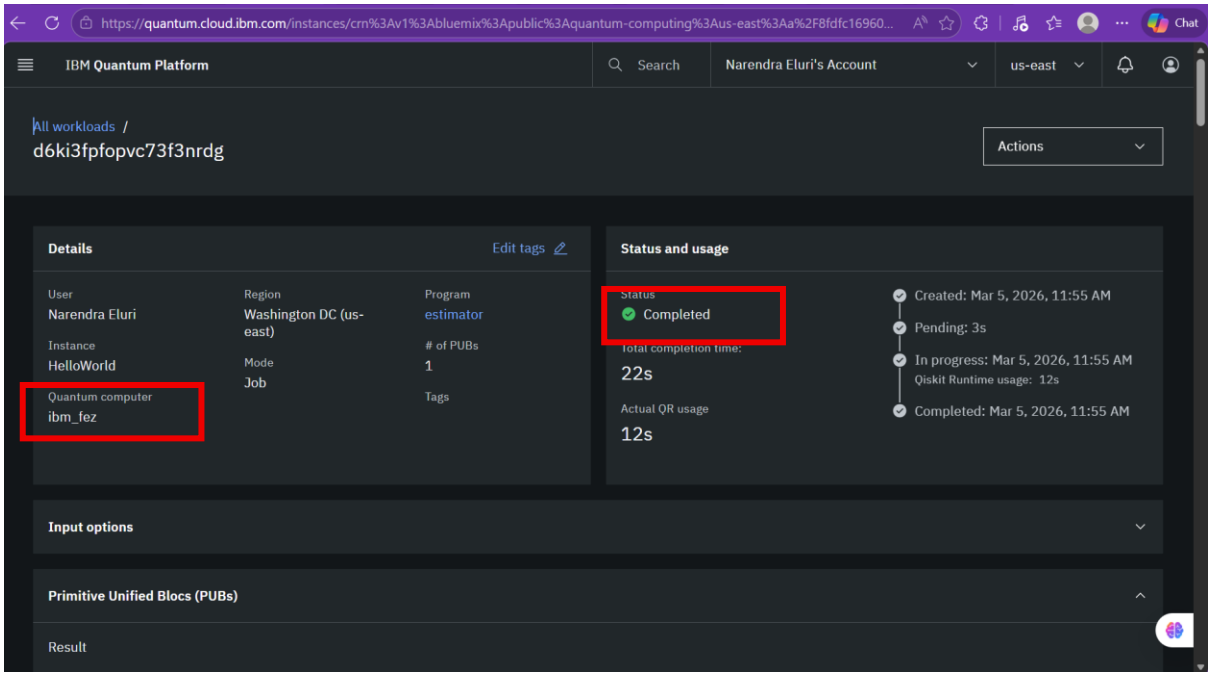
```
[ ]
from qiskit_ibm_runtime import QiskitRuntimeService

QiskitRuntimeService.save_account(
token="[redacted]", # Use the 44-character API_KEY you created and saved from the IBM Quantum Platform Hc
instance="c[redacted]:", # Opt
overwrite=True
)
```

16. Run all the cells
17. Your Jupyter notebook will be executed by selecting real time Quantum hardware from IBM Quantum Platform.
18. Now switch to the IBM Quantum platform
19. The executed quantum circuits will appear in the **My Recent Workloads** section of the IBM Quantum Platform, where each circuit execution is recorded as a **Job** with its status and execution details.



20. Click on the **Job ID** to view detailed information about the executed job, including the name of the quantum processor (QPU) used for real-time execution, pending time etc.





21. **Congratulations** You are first Quantum circuit is executed on the real time Quantum Processor.